Replication of the polytomous person-fit Monte Carlo in Beck (2023)

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Abstract

Do the classification rates and the ANOVA tables in Beck (2023) follow from the Monte Carlo described there? Beck simulates six response biases on ordered items with four, five, six, or seven options, one to four dimensions, and tests of 12 or 36 items, scores $U3$, normed Guttman errors, and H_i^T against a bootstrapped 5th-percentile cutoff, and reports sensitivities from 0 to 0.29 with specificities from 0.94 to 0.98. The partial omega squared values in Appendix C were recomputed from the printed F statistics at $N = 32{,}000$, and the four rates in Tables A24–A26 were checked against the identity that holds when 50 of 1,000 respondents are flagged. All 504 omega squared values agree with that formula to within 0.005. Specificity and negative predictive value in the three grand tables sit within 0.006 of the identity. A 40-replication rerun under the linear predictor printed in Equation 23, with the extreme-response constant set to 0.5, does not recover the sensitivity levels. The nearest cells are normed Guttman errors for midpoint responding, 0.0035 against a printed 0.00, and for extreme responding, 0.284 against 0.27. Both sit outside a two-standard-error band, and a Holm correction on the eighteen grand sensitivities leaves the misses in place.

1 The question, and the headline

This section answers whether the printed rates and the printed ANOVA tables come from the Monte Carlo in the dissertation. The ANOVA arithmetic and the relationship among the four rates do. The sensitivity levels do not, under the assumptions stated below.

Beck (2023), https://digitalcommons.unl.edu/cehsdiss/425, studies whether three nonparametric person-fit statistics detect response bias on ordered polytomous items. The design is six bias types, four numbers of response options, four dimensionalities, and two test lengths: 192 conditions. Each condition has 900 ordinary respondents and 100 biased respondents, repeated 1,000 times. The statistics are $U3$, the normed number of Guttman errors, and the person coefficient H_i^T. The cutoff in each replication is the median of the 5th percentiles of 1,000 bootstrap samples. Tables A24–A26 collect the rates. Sensitivities run from 0.00, for midpoint responding on the normed Guttman errors and on $U3$, to 0.29, for careless responding on the normed Guttman errors. Specificities run from 0.94 to 0.98.

Two checks use only the printed numbers. Partial omega squared, computed from each printed F at $N = 32{,}000$, matches all 504 entries in Tables C1–C18 to within 0.005, which is the half-unit of the printed two-decimal column. When exactly 50 of 1,000 people are flagged and 100 of them are the biased respondents, specificity, positive predictive value, and negative predictive value are functions of sensitivity. Specificity and negative predictive value in Tables A24–A26 sit within 0.006 of that identity. Positive predictive value sits within 0.02.

The generative rerun does not land on the sensitivity column. The reported run uses the linear predictor in Equation 23 as printed, 40 replications, and an extreme-response constant $\varepsilon = 0.5$. Of the eighteen grand sensitivities, none fall inside a two-standard-error band, before or after Holm. The normed Guttman errors come nearest: midpoint responding is 0.0035 against 0.00, and extreme responding is 0.284 against 0.27. The combined standard errors are 0.00018 and 0.00088. Two factor-margin cells do match, and two more sit just outside the band. $U3$, computed from the same definition as PerFit 1.4.7, does not follow the printed $U3$ column on either tail.

The original work is the design, the bias definitions, and the printed tables. The calculations below are ours: the arithmetic checks, the Monte Carlo, and the comparison. Multiplicity is applied only to our check that a printed rate and our rate agree. The dissertation's own tests are one classification per replication.

2 What we replicated

This section answers which printed claims agree with a direct calculation. Table 1 is that comparison. "Match" means the same number, or a Monte Carlo gap no larger than two combined standard errors. "Partial" means one piece of the claim holds and another does not. "Close" means a gap sitting on the edge of that band, or a gap that disappears at the printed rounding. "No match" means a rate that remains outside the band. The combined standard error uses the printed count of 1,000 replications and our count of 40, with our within-cell standard deviation as the plug-in for both.

The omega squared row is an algebraic check, not a second simulation. For an effect with degrees of freedom ν and printed F,

$$\hat{\omega}_p^2 = \frac{\nu(F-1)}{\nu(F-1)+N}, \qquad N = 32{,}000.$$

The largest absolute gap across the 504 effects is 0.0050. The script is `simulation/extract_published.py`, and the row-level file is `simulation/out/anova_check.csv`.

The identity row uses a fixed flag count. With 100 biased respondents and 50 flags,

$$\mathrm{TP} = 100\,s, \quad \mathrm{FP} = 50 - \mathrm{TP}, \quad \mathrm{spec} = \frac{900-\mathrm{FP}}{900}, \quad \mathrm{PPV} = \frac{\mathrm{TP}}{50}, \quad \mathrm{NPV} = \frac{850+\mathrm{TP}}{950},$$

where s is sensitivity. A 5% rule cannot produce a sensitivity above 0.50. Every printed grand sensitivity is at most 0.29, so the identity is inside that bound. The printed rows are averages of ratios, so the identity applied to the averaged sensitivity is an approximation. On the 18 grand rows other than the mean row, all three derived rates sit within 0.02 of the printed rates, and specificity and negative predictive value sit within 0.006. The file is `simulation/out/identity_residuals.csv`.

Table 2 is the normed-Guttman grand column at $\varepsilon = 0.5$. Midpoint responding is the floor in the dissertation and in the rerun. Extreme responding is the highest rate in that column once careless responding is set aside, in the dissertation and in the rerun. The levels still miss the two-standard-error band. Careless responding sits on the 0.50 ceiling, which is what perfect separation looks like when 50 flags are available for 100 biased respondents. The printed careless rate is 0.29.

The midpoint grand rate prints as 0.00 at two decimals, and 0.0035 also prints as 0.00. Against an exact zero the gap is about twenty standard errors, so the row stays a Monte Carlo miss, and it is also a rounding display. The same remark covers the factor-margin match at seven options, 0.0031 against 0.0026.

Claim	Published	Ours	Verdict
Partial omega squared, 504 effects	Two decimals in Tables C1–C18	From the printed F, within 0.005	match
Grand specificity and NPV	Tables A24–A26	Within 0.006 of the 5% identity	match
Grand positive predictive value	Tables A24–A26	Within 0.02 of that identity	close
Guttman, acquiescence, length 36	0.072	0.0718	match
Guttman, midpoint, seven options	0.0031	0.0026	match
Guttman, extreme, three dimensions	0.260	0.263	close
Guttman, extreme, six options	0.281	0.275	close
Guttman grand, midpoint	0.00	0.0035	no match
Guttman grand, extreme	0.27	0.284	no match
Guttman grand, careless, every item	0.29	0.500	no match
Eighteen grand sensitivities	Tables A24–A26	0 of 18 inside two SE; Holm keeps the misses	no match
$U3$ grand column, either tail	Same shape as the Guttman column	The two columns separate	no match

Table 1: Printed claims against our calculations. Rates are proportions. The Monte Carlo rows use $\varepsilon = 0.5$ and the aberrant-direction tail: low H_i^T, high normed Guttman errors, high $U3$.

H_i^T is higher than the printed column on five of the six biases. Disacquiescence is the nearest, 0.033 against 0.04, with combined standard error 0.00049. $U3$ on the upper tail flags midpoint responding at 0.34 where the page prints 0.00, and flags acquiescence at 0.51 where the page prints 0.05. The lower tail of $U3$ matches the printed midpoint zero in the $\varepsilon = 1$ run and misses it, at 0.0048, in the $\varepsilon = 0.5$ run. The rest of that lower-tail column does not follow the page.

Of the 60 factor-margin sensitivities for the normed Guttman errors at $\varepsilon = 0.5$, two are inside the raw band and two are close. After Holm, three of the sixty are retained as matches. The two raw matches are Table A6, test length 36, acquiescence, and Table A12, seven options, midpoint responding. The full grid is `simulation/out/comparison_eps05.csv`.

3 What we did ourselves, and what we took from others

This section answers which pieces are ours. The dissertation supplies the design, the equations, Tables A1–A3, and the result tables. We supply the Monte Carlo, the arithmetic checks, and the comparison.

Responses are drawn from `mirt` 1.36 [2] through `simulation/monte_carlo.R`. The person coefficient is the rearrangement formula in the `se = FALSE` branch of `mokken` 3.1.2 [3], written out in `simulation/personfit.R`. Normed Guttman errors and $U3$ follow the complete-data branches of PerFit 1.4.7 [4], in `simulation/personfit.R` and `simulation/personfit.R`. On six random

Bias	Published	Ours	SE	Verdict
Acquiescence	0.05	0.062	0.00055	no match
Disacquiescence	0.13	0.047	0.00055	no match
Midpoint	0.00	0.0035	0.00018	no match
Extreme	0.27	0.284	0.00088	no match
Social desirability	0.08	0.031	0.00042	no match
Careless	0.29	0.500	0.00005	no match

Table 2: Normed Guttman sensitivity, Tables A24, from 40 replications at $\varepsilon = 0.5$. The standard error is the combined standard error of the two means.

matrices the person coefficient matches `coefH` with a maximum absolute difference of 0. The other two statistics match the package before it rounds the exported scores to four decimals; the largest absolute difference against those rounded scores is 5×10^{-5}. That check is `simulation/out/validate_personfit.csv`. The graded probabilities from `mirt` match plogis$(a^\top\theta - b)$ with a maximum absolute difference of 0 in one and two factors.

R 4.3.3 [1] is the language because the dissertation names R. `mirt` is the generator the dissertation names. The current `mirt` asks for `Deriv`, and the current `Deriv` asks for R 4.5, so the run uses `mirt` 1.36 (21 March 2022) with `Deriv` 4.1.6 from the CRAN archive. `mokken` and PerFit are the scoring packages the dissertation names. MASS 7.3-60.0.1 [5] draws the latent normal vectors. Versions are in `simulation/out/session_info.txt`.

4 Assumptions

This section answers which inputs we supplied because the dissertation does not print them. Each subsection is one such input. A choice the dissertation states is recorded in Section 5.

4.1 The size of the extreme-response constant

Equation 26 prints ε as a constant and gives its sign. The magnitude is not printed. The reported run uses $\varepsilon = 0.5$, so midpoint boundaries are spread by $e^{0.5}$ and extreme boundaries are shrunk by $e^{-0.5}$. The same grid at $\varepsilon = 1$ gives an extreme-response Guttman sensitivity of 0.186. At $\varepsilon = 2$ that sensitivity is 0.074. At $\varepsilon = 0.5$ it is 0.284, against the printed 0.27. Midpoint sensitivity stays at the floor for all three values: 0.0035, 0.0017, and 0.000. The shift biases do not use ε, and their rates agree across the three runs to three decimals. The files are `simulation/out/mc_ink_eps0.5_c1_r40.csv`, `simulation/out/mc_ink_eps1_c1_r40.csv`, and `simulation/out/mc_ink_eps2_c1_r8.csv`.

4.2 Seven-option boundaries

Table A2, Figure 1, prints uniforms for four, five, and six options and has no row for seven. The prose names terciles, quartiles, and quintiles and stops at six options. Seven options need six boundaries. The run splits the widest six-option interval, from -13.71 to -6.68, at its midpoint, and uses the six contiguous intervals that produces. That choice is `simulation/monte_carlo.R`.

Uniform Distributions of Item Boundary Parameters by Condition

k	Cat. 1	Cat. 2	Cat. 3	Cat. 4
4	*U*(-13.71, -5.19)	*U*(-5.19, -2.51)	*U*(-2.51, 2.42)	
5	*U*(-13.71, -5.81)	*U*(-5.81, -2.19)	*U*(-2.19, 2.42)	*U*(1.5, 3)
6	*U*(-13.71, -6.68)	*U*(-6.68, -4.78)	*U*(-4.78, -3.09)	*U*(-3.09, -1.42)

Note. Cat stands for Category

Figure 1: Table A2. The $k = 7$ row is absent. For $k = 5$, category 3 and category 4 overlap.

4.3 One draw of the boundaries, then a sort

Table A2 gives distributions. The realized boundaries are not printed, and no seed is printed. The run draws each boundary once from its interval under seed 20230511, then sorts the boundaries within each item so they increase. For four and six options the intervals share only endpoints, so the sort does not move an interior draw. For five options, category 3 is $U(-2.19, 2.42)$ and category 4 is $U(1.5, 3)$, so independent draws can arrive out of order. The item indices in each of the 32 static tests are `simulation/out/item_samples.csv`.

4.4 Careless replacement of every item

The dissertation says a uniform random option replaces the raw response on the selected items, and that each option is equally likely. It does not say how many items are selected. The reported run replaces every item. Replacing a fraction 0.5, 0.2, or 0.1 of the items leaves the careless Guttman sensitivity at 0.500, 0.485, and 0.444, against the printed 0.29. The gap does not close under those fractions. The files are the `c0.5`, `c0.2`, and `c0.1` tables in `simulation/out`.

4.5 The sample standard deviation in the shift

Equations 24 and 25 add or subtract s_{δ_i}, the standard deviation of item i's boundaries. The run uses the sample standard deviation, with divisor $n - 1$. Figure 2 is the printed shift.

$$\delta^{*}_{ki} = \delta_{ki} + s_{\delta_i}, \tag{24}$$

Figure 2: Equation 24, the acquiescence shift.

4.6 Scaling constant D = 1

The paragraph under Equation 23 calls $D = 1.702$ optional and $D = 1$ the other choice. The run uses $D = 1$, which is the logistic metric of `mirt`'s graded generator.

4.7 One latent sample, then a trim to 900

The dissertation draws 1,000 latent vectors, drops zero-variance response rows, and then drops rows at random until 900 remain. The run draws one shared sample of 1,000 from the Table A1 normal distribution and trims each of the 32 response matrices separately. The means and covariances are Figure 3. The socially desirable means are the four printed values in Figure 4, with the same covariance.

Covariance Matrix and Means for Data Simulation

	Mean	**D1**	**D2**	**D3**	**D4**
D1	*0.03*	1.02			
D2	*0*	0.49	1.11		
D3	*0.09*	0.31	0.38	0.95	
D4	*0.02*	0.43	0.89	0.40	0.96

Figure 3: Table A1. The matrix is positive definite, with eigenvalues 2.52, 0.71, 0.67, and 0.14.

standard deviation below the latent means used to create the base data sets (the latent means will be: ξ_1 = -.982; ξ_2 = -.964; ξ_3 = -.903; and ξ_4 = -.868). All other aspects of this multivariate normal distribution were kept the same as the distribution used to generate

Figure 4: The socially desirable means on page 66.

4.8 A sample quantile in place of the bootstrap median

The dissertation takes the median of 1,000 bootstrap 5th percentiles. The run uses R's type-7 sample quantile at 0.05 or 0.95. On one continuous sample of size 1,000, that median and the sample 5th percentile differ by 3×10^{-4}. On a sample with a point mass, they differ by 6×10^{-4}. The file is `simulation/out/bootstrap_cutoff.csv`, from `simulation/bootstrap_cutoff.R`. The comparison bands are an order of magnitude larger than those gaps only for the careless ceiling, where the standard error is 5×10^{-5}. Elsewhere the quantile substitution is smaller than the band and does not account for the misses.

5 Ambiguities that were resolved

This section answers which readings of the page were settled by an artifact. Each subsection names the script, the output, and the command. The commit that contains them is the commit that added this note. Regenerate with the commands in `README.md`.

5.1 Equation 23 is the sum that multiplies the threshold by each slope

Figure 5 prints

$$P_{jk}(\theta) = \frac{1}{1 + \exp\left[-D \sum_{h=1}^{H} a_{jh}(\theta_h - b_{jk})\right]}.$$

The usual graded predictor, and `mirt`'s, is $a^\top\theta - b_{jk}$. The printed sum equals $a^\top\theta - b_{jk}\sum_h a_{jh}$. The rerun passes `mirt` the intercept $-b_{jk}\sum_h a_{jh}$. Under the usual intercept $-b_{jk}$, the trim to 900 non-constant rows fails on the grid: a direct run stopped with fewer than 900 such rows in the cells it reached. A single twelve-item slice under that usual intercept has 703 non-constant rows out of 1,000. The same slice under the printed sum has 754. The full printed-sum grid does trim to 900 in every cell. The script is `simulation/standard_collapse.R`. The generator is `simulation/monte_carlo.R`.

2016). One expression of the MGRM is given in Equation 23:

$$P_{jk}(\boldsymbol{\theta}) = \frac{1}{1 + \exp\left[-D \sum_{h=1}^{H} \left[a_{jh}\left(\theta_h - b_{jk}\right)\right]\right]}, \tag{23}$$

Figure 5: Equation 23 as printed.

5.2 The aberrant tail is the high tail for U3 and for Guttman errors

PerFit documents both statistics as increasing in misfit, from 0 to 1. H_i^T is low when a person is less scalable. The method chapter states one 5th-percentile rule for all three. Flagging the low tail of the normed Guttman errors, where many respondents share a value at the quantile, flags about 283 people rather than about 50, and the careless sensitivity on that tail is 0. Flagging the high tail flags about 51 people. The high tail is the reading in Table 2. The low tail is in the same output file, under the column prefix `G_lower`.

5.3 Table A1 is the lower triangle including the variances

The prose says the off-diagonal covariances run from 0.31 to 1.11. Figure 3 places 1.11 on the diagonal of the second dimension, and places 0.49, 0.31, 0.38, 0.43, 0.89, and 0.40 off the diagonal. The simulation uses the figure. The prose on the same page says the means run from -0.001 to 0.092. The figure prints 0, 0.02, 0.03, and 0.09. The simulation uses the figure.

5.4 The package scores match the fast formulas

`mokken::coefH` with standard errors on a $36 \times 1{,}000$ matrix exhausted memory in this environment. The `se = FALSE` branch is a covariance and a covariance of the column-sorted matrix, and it finishes in a fraction of a second. The validation command is `Rscript validate_personfit.R out`. The output is `simulation/out/validate_personfit.csv`.

6 What remains unresolved

This section answers which gaps are still open, and which variations were already run.

The eighteen grand sensitivities remain outside the two-standard-error band after Holm. The normed Guttman column has the same shape as the page: a floor at midpoint responding, a high rate at extreme responding, and a ceiling at careless responding. The levels differ. $\varepsilon \in \{0.5, 1, 2\}$ moves the extreme rate through 0.284, 0.186, and 0.074, against 0.27. The value 0.5 is the closest of those three and is still 0.014 above the page, with a standard error of 0.00088. Careless fractions 1, 0.5, 0.2, and 0.1 leave that sensitivity at 0.500, 0.500, 0.485, and 0.444, against 0.29. The fraction does not close the gap under this boundary draw.

$U3$ remains a separate miss. The statistic matches PerFit 1.4.7 before rounding. On these replicates it does not track the printed $U3$ column, which tracks the printed Guttman column. A different PerFit release from 2023, or a different boundary draw, could still produce that tracking. This run does not.

The realized boundaries, the item sample, and the seed are unprinted. A different draw from Table A2 can move a grand mean by more than the Monte Carlo standard error of 1,000 replications, because that standard error describes the redraw of the 100 biased respondents with the items held

fixed. Our band uses our within-cell dispersion as a stand-in for theirs. It is the right band for two runs of one fixed item set. It is a lower bound on the uncertainty that comes from redrawing the items. The misses are large beside that lower bound: several hundredths, on bands of about a thousandth. Redrawing the items remains a live explanation, and so does the unprinted ε.

The sentence that the socially desirable means lie one standard deviation below Table A1 does not recover -0.982, -0.964, -0.903, and -0.868. The run uses those four printed numbers. Beck (2015), the master's thesis cited as the source of Table A1 and of the ranges behind Tables A2 and A3, was not retrieved.

The bootstrap-median cutoff was checked on two artificial samples, not on every replication of the grid. The differences were 3×10^{-4} and 6×10^{-4}.

7 Code and source availability

This section answers where the inputs and the outputs live. The dated ledger is in `INVESTIGATION.md`.

The dissertation PDF used here has sha256 `86c3bb482efadc385cd52f7ea12dc33c50b9b70b4ce6111aedb075a12968f62b`. It is the Wayback snapshot `20240907085051` of `https://digitalcommons.unl.edu/cgi/viewcontent.cgi?article=1436&context=cehsdiss`. A direct request for that URL on 28 October 2026 returned a Cloudflare challenge page. The PDF is not in git.

No sentence in the dissertation names a script archive. The public GitHub account `MFBeck` has two repositories, `Hello_Jenkins` and `jenkins_test`, cloned on 28 October 2026; both are Jenkins demonstrations. An OSF search, a Zenodo search, and a GitLab search on that date did not return a simulation archive for this dissertation. PerFit's source for the two polytomous statistics is the repository `https://github.com/jorgetendeiro/PerFit`, version 1.4.7.

8 Next steps

This section answers what would replace an assumption with a printed input. The Beck (2015) thesis, ProQuest 1588285, is the cited source of the means, the covariance, and the boundary ranges. A seed, a boundary matrix, or an R `sessionInfo` record from the dissertation run would decide whether the sensitivity gaps are the unpublished draw. The check is the same Monte Carlo with those inputs in place of seed 20230511 and of $\varepsilon = 0.5$.

9 Artifacts

This section lists each claim, the file that holds it, and the command that regenerates it. Commands run from `formalization/polytomous-person-fit/simulation` unless the path says otherwise. The PDF path below is the copy named in Section 7.

Claim	File	Command
Discriminations, Tables A6–A26, Tables C1–C18	simulation/out/discriminations.csv and the published CSV files	`python3 extract_published.py PDF out`
Omega squared from the printed F	simulation/out/anova_check.csv	the same command
Identity residuals	simulation/out/identity_residuals.csv	the same command
Package agreement	simulation/out/validate_personfit.csv	`Rscript validate_personfit.R out`
Forty replications, $\varepsilon = 0.5$	simulation/out/mc_ink_eps0.5_c1_r40.csv	`Rscript monte_carlo.R out 40 ink 0.5 4 1`
Forty replications, $\varepsilon = 1$	simulation/out/mc_ink_eps1_c1_r40.csv	`Rscript monte_carlo.R out 40 ink 1 4 1`
Comparison, $\varepsilon = 0.5$	simulation/out/comparison_eps05.csv	`Rscript score_gaps.R` on that file
Quantile against bootstrap median	simulation/out/bootstrap_cutoff.csv	`Rscript bootstrap_cutoff.R out`
Crops	paper/figures	`python3 paper/figures/crop_pages.py PDF`

References

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